

Agha Aghayev

 [aghayevagha.github.io](https://github.com/aghayevagha) |  [aghayevagha](https://github.com/aghayevagha) |  [LinkedIn](#) |  [email](#) |  [scholar](#)
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Experience

Research Assistant

May 2025 – Present

Azerbaijan Technical University, Baku, Azerbaijan

Supervisor: Dr. Yadigar Imamverdiyev

- Assisting in research projects focused on cryptography.

Adjunct Instructor

September 2025 – Present

Baku Higher Oil school, Baku, Azerbaijan

Courses

- Advanced Cryptography and Data Protection (graduate level)
- Discrete Mathematics (undergraduate level)

Junior Machine Learning Engineer

Dec 2023 – July 2024

PRODATA

- Smart agriculture solutions for satellite imagery using machine learning.
- Implemented algorithms for water detection and volume estimation.

Publications

1. **A. Aghayev.** "Implementation of Lattice Trapdoors for Quantum-Proof Identity-Based Encryption," 2025 6th International Conference on Problems of Cybernetics and Informatics (PCI), Baku, Azerbaijan, 2025, pp. 1-5, doi: 10.1109/PCI66488.2025.11219723. ([link](#))
2. **Aghayev, A.**, Imamverdiyev, Y., & Orujlu, O. (2025). A REVIEW OF CRYPTANALYSIS TECHNIQUES FOR LATTICE CRYPTOGRAPHY USING MACHINE LEARNING. Reliability: Theory & Applications, 20(SI 10 (88)), 176-185. ([link](#))
3. **Agha Aghayev** and Yadigar Imamverdiyev. Privacy-preserving Shape Matching with Leveled Homomorphic Encryption ([Preprint](#))
4. **Agha Aghayev** and Nour-edine Rahmani. Cryptanalysis of a Post-Quantum Signature Scheme Based on Number-Theoretic Assumptions ([Preprint](#))

Projects

Lattice-based trapdoor sampling ([github](#))

- Implementation of Discrete Gaussian sampling for the GPV scheme with full-rank bases and Gadget-based trapdoors (with perturbation sampling)
- Built quantum-resistant public key encryption, such as Dual-Regev scheme.

Privacy-Preserving Shape matching with OpenFHE ([github](#))

- Utilized leveled CKKS scheme for secure shape matching
- Approximation of non-linear functions with Chebyshev's interpolation method

Grobner basis attacks on multiple-key NTRU problem ([github](#))

- Arora-Ge modeling for converting NTRU learning instances to error-free multi-variable polynomial system of equations
- Solving system with Grobner basis computation and investigating speed-ups

Seminars

Lattice-based cryptography

3 December, 2025

I presented core topics on lattice-based cryptography, LWE and SIS problems, their cryptographic applications and relation to worst-case lattice problems, algebraic attacks on LWE, etc.

Education

MS in Computer Science and Data Analytics (Diploma of Honors)

GWU, Washington DC, United States (3.79 / 4) / ADAU, Baku, Azerbaijan (3.81 / 4).

Sep. 2023 – Jun. 2025

Thesis: *Lattice-based Identity-based encryption: Foundations, Constructions, and Implementation* under supervision of Prof. Arkady Yerukhimovich. ([pdf](#))

BS in Computer Science (Diploma of Honors) - GPA : 92.83/100

ASOIU, Baku, Azerbaijan.

Sep. 2019 – Jul. 2023

Technical Skills

Languages: Sagemath, C++, Python, Java, Haskell, L^AT_EX, HTML and CSS (basic)

Developer Tools: Git, Linux

Frameworks: OpenFHE, Tenseal, PyTorch, Spring